Tyler Nielsen

The Impact of Artificial Intelligence

Artificial intelligence is a topic that has been gaining traction in recent years, and for good reason. Many advancements have been made in this area that have put artificial intelligence under the spotlight. While artificial intelligence could have a revolutionary impact on a variety of industries and society as a whole as it continues to develop and improve, it is unclear exactly how it will impact society at large. Many people are concerned that artificial intelligence will one day take their jobs, while others are excited about the concept of it taking a load off of us humans. Many people are worried that artificial intelligence will enhance the sophistication of cyber attacks, while others are excited about its potential for enhancing security. There are countless areas that artificial intelligence can both benefit and harm to varying capacities. It is hard to predict the scale of this, and at this point the technology is likely too far along to halt the continued development of it. However, it might be time to start worrying about the ways artificial intelligence can negatively impact both individuals and society as a whole.

The first and most obvious way that artificial intelligence will impact society is in the job market. Artificial intelligence has the potential to revolutionize a variety of fields, but what comes with that is the potential for job displacement. Some people think of a utopian future of AI, where it does the vast majority or all of the work. However, not only is that unrealistic for the near future, getting to that point will involve a lot of hardships along the way. The only way this future could be achieved is through different jobs being slowly faded away as AI starts to take them. It is impossible for artificial intelligence to instantaneously develop to the point where it can do all jobs, meaning that people whose fields are taken over, especially in the earlier stages of AI development, will simply be out of a job. According to IBM, 35% of companies are already reported to be using AI in their business, and 42% are reported to be experimenting with it (IBM Global AI Adoption Index, pg. 3). Corporations tend to only care about the bottom line, they wouldn't want to incorporate AI into their tasks if they didn't believe it would save them money. While there are more ways AI can save money than just replacing human workers, that is definitely a consideration they have in regards to improving profits. In fact, according to The Challenger Report of May 2023, 3,900 jobs were lost due to the direct impact of AI in May of 2023. Considering how AI is still in its relatively early stages of development, that number is actually quite staggering. Of course, we aren't at a stage where all that many jobs can be entirely automated by artificial intelligence, but it absolutely is at a stage where it can increase

the productivity of any given individual employee. Suppose utilizing AI can make someone twice as productive in their work, the company may only need half as many employees for that job. While it is also entirely possible and likely that artificial intelligence will create entirely new jobs, there is no way that the number of jobs created by AI will outweigh the number taken by it. To see a similar instance in action, look no further than self-checkout. “In 2019, a report found that 75,000 retail jobs were lost to self-service tills and other automation” (Reed). In the case of self-checkout, jobs were absolutely created in the fields of developing and maintaining self-checkout systems, but it is absolutely unrealistic to suggest that it created anywhere near 75,000 jobs. Artificial intelligence will almost certainly mirror this outcome. While artificial intelligence holds the promise of revolutionizing various industries, the impact that will have on the job market is undeniable.

A noticable issue is that you make dire predictions w/out positive predictions to balance it.

Another area that artificial intelligence is bound to impact is the realm of cybersecurity. Artificial intelligence is already being used in both the offense and defense of cybersecurity systems. However, even without the use of artificial intelligence in constructing an attack, the use of AI to defend a system may introduce new vulnerabilities that people do not predict. According to NIST, there’s a variety of ways that artificial intelligence can be manipulated in cyber attacks through means of altering how the AI responds to certain inputs and/or corrupting the training data used by the AI (nist.gov). If an attacker manages to manipulate the data used to train an AI defense system, it could create an array of vulnerabilities in that system that the attacker will have full knowledge of. If this type of attack goes under the radar, it could be devastating for the system being attacked. Of course, measures will be put in place to prevent this type of thing from happening, but if it does happen, people may not be ready for it. Again, that isn’t even considering the use of AI to actually construct an attack. Artificial intelligence could enable unskilled individuals to perform sophisticated cyber attacks. This may increase both the number of attacks and the variety of attacks, potentially overwhelming defense systems. This idea has been explored by the National Center for State Courts, in which they state that “by lowering the barrier of entry to novice cyber criminals, hackers-for-hire and hacktivists, AI enables relatively unskilled threat actors to carry out more effective access and information-gathering operations” (NCSC). They go on to state that cyber criminals have already started developing criminal generative AI. Generative AI can be used in several types of cyber attacks, such as social engineering and malware generation. AI can be used to improve the believability of social engineering attacks such as phishing. “Generative AI can make traditional phishing attacks - via emails, direct messages and spurious websites - more realistic by eliminating spelling errors and grammatical mistakes and adopting convincingly professional

writing styles" (Krishnan). At the current time, so many phishing attacks can be immediately disregarded just by reading a sentence or two and seeing the absurd grammatical mistakes. With the use of artificial intelligence to produce phishing emails and other social engineering techniques, that method of detection could be entirely eliminated. And as phishing emails become more and more believable with the use of artificial intelligence, they become more and more likely to get through email filters meant to block users from ever even seeing them. Artificial intelligence can also be implemented to detect these more sophisticated phishing attacks, however, if artificial intelligence reaches the point where these emails are indistinguishable from genuine looking human written ones, it may be impossible for even AI systems to detect them. The article by Krishnan also mentions how AI can be used to incorporate real time information into phishing emails at rates faster than humans could ever replicate, further improving their believability. The number one vulnerability in any business or security system is human error. Artificial intelligence based attacks will absolutely be targeting that in ways more effective than ever before, especially with the development of deep fakes and similar technologies. The increased sophistication of attacks through the use of artificial intelligence poses unprecedented challenges in cybersecurity, and handling these challenges is a task that may overwhelm traditional security paradigms and require a rethinking of defensive approaches.

Artificial intelligence is also going to contribute to the difficulty in distinguishing what content is real and what content is fake. The development of deep fake technology is going to enable people to create believable fake images and videos in a matter of minutes or even seconds. This will contribute to the spread of misinformation and even has the potential to ruin lives. "Deepfakes can depict individuals in compromising or harmful situations, potentially leading to significant reputational damage. The creation and dissemination of deepfakes can cause considerable emotional distress to the depicted individuals, especially if the content is defamatory, embarrassing, or violates their personal or professional ethics" (Jodka). The rise of deep fakes being used to defame and slander people is a serious risk involved with AI technology. Fake videos that are damaging to someone's reputation could ruin their life and get them fired from their job. Even if the slanderous content is proven to be fake, the damage it caused may already be done. The deep fake content could even be used to persecute someone for a crime they didn't commit. It could also be used to allow real criminals to get away with heinous crimes, on the grounds that a video of the crime is just a deep fake, even if it wasn't. This technology completely blurs the line between what is real and what is fake. Even if AI systems are put in place to detect deep fakes to prevent people from being criminally

prosecuted on the grounds of a faked image or video, such technology likely will not be developed until it is already a problem, and even then the deep fakes can be improved to circumvent detection. “Deep fake detection technology is in the works by institutions like MIT, Northwestern and even OpenAI itself, but the cat-and-mouse game of development versus detection will likely continue” (Curry). Even if a perfect deep fake detection system is put in place, it would make it too late for those already prosecuted for crimes they didn’t commit on the grounds of deep fake video evidence. Curry’s article also brings up how image and video evidence used to be the gold standard of court evidence. If you had a video of a crime, at that point it should be case closed, but with AI deep fakes, that will no longer be the case. While the risks associated with deep fake technology are significant, it’s important to recognize its potential positive uses, for example in the realm of creative works. “It has been speculated that deepfakes can and will be used to customize and personalize movies, video games, and other media by superimposing one’s face onto the characters” (Murphy). The development of deep fakes for purposes like this are fine, but when the same technology can be used to ruin lives and spread misinformation, it raises the question of if it is actually worth it. Why develop technology that can ruin lives just because it can help make movies? Another common argument in favor of deep fakes is how they can be in a classroom. “Deepfake algorithms can animate historical photos and footage, allowing influential figures to give speeches and presentations as if they were in the classroom. The resulting videos are far more engaging and interactive than lectures or textbooks” (Patterson). While this use of deep fakes in this way is an interesting concept, is it really necessary? Students can learn history without needing to see deep fake animated historical figures. Again, these minor benefits that deep fakes can introduce are simply not worth the devastation that the same technology can cause.

In conclusion, the development of artificial intelligence may be inevitable, but the impact it will have on society is likely to be negative before it can be good. Artificial intelligence will introduce problems such as job displacement, increased rates and effectiveness of cyber crimes, and the spread of believable misinformation through deep fakes that could lead to defamation and miscarriages of justice. While these issues are some of the most serious, the negative impacts artificial intelligence may have far exceed the scope of what we can currently predict. Even in light of the ways artificial intelligence may benefit society in the long run, it seems inevitable that there will be serious consequences for the development of AI technology. It is crucial that people understand these risks and are prepared for them. Being knowledgeable of how artificial intelligence can improve phishing emails and spread misinformation will help

mitigate these risks associated with the technology. As artificial intelligence continues to develop, society needs to be prepared for all of the ways it will change our world as we know it.

Good job, but it seems like 3 seperate papers pg. breaks?

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